DECK 04 · WEEK 2 · MORNING 2

Microeconomics & Business Cycles

How firms set price and output, and how the economy cycles around its potential path

CFA Level I · Vol 2 Learning Modules 1–2

Roadmap — Microeconomics & Business Cycles

6 sections · ~2 hours · Built bottom-up from CFA readings

01	**Demand, supply, elasticity**	Law of demand · own-price / cross-price / income elasticity · revenue reaction
02	**Firm costs & profit maximization**	Economic vs accounting profit · MR = MC · breakeven / shutdown · economies of scale
03	**Four market structures**	Perfect / monopolistic / oligopoly / monopoly · Cournot duopoly · Nash · HHI
04	**Business cycle phases & indicators**	Four phases · classical vs growth · credit cycles · LEI / coincident / lagging
05	**Unemployment & inflation**	Labor-force measures · CPI · core · PPI · GDP deflator · Phillips curve · Okun's law
06	**Investment-relevant cycle dynamics**	Leading markets · inventory and capex cycles · sector rotation playbook

Learning objectives

By the end of this deck you should be able to:

1. Compute and interpret own-price elasticity of demand, and predict how total revenue will move when a firm changes price given the elasticity regime.

2. Apply the MR = MC profit-maximization rule to any market structure, and distinguish the breakeven, shutdown, and long-run-exit thresholds.

3. Identify which of the four market structures a firm operates in using product differentiation, number of sellers, barriers to entry, and pricing power.

4. Describe the four business-cycle phases and translate inventory, capex, employment, and credit indicators into a phase call.

5. Link unemployment and inflation through the short-run Phillips curve and Okun's law, and translate an unemployment forecast into an inflation forecast.

SECTION 01

Demand, supply, elasticity

Demand slopes down and supply slopes up because buyers face diminishing marginal utility while sellers face rising marginal cost

Market equilibrium is the (P, Q*) pair at which the quantity buyers demand equals the quantity sellers supply — the starting point for every subsequent structure.*

At equilibrium: Q_D(P*) = Q_S(P*) → no surplus, no shortage

where: Shifts in either curve move P and Q* in predictable directions; elasticity determines which moves more.*

LAW OF DEMAND

Holding income, prices of related goods, and preferences constant, higher P ⇒ lower Q_D. Two drivers: substitution effect (relative price up) and income effect (real purchasing power down).

LAW OF SUPPLY

Holding input prices and technology constant, higher P ⇒higher Q_S. Driver: rising marginal cost beyond some threshold — producers need a higher price to cover MC on the next unit.

SHIFTS vs MOVEMENTS

A price change MOVES along the curve; a change in income, input prices, or technology SHIFTS the curve. Students confuse this on day one — do not.

Source: CFA Institute, Curriculum Vol 2, Learning Module 1, Introduction, pp. 3–7.

Own-price elasticity measures how responsive quantity demanded is to price — and it determines whether cutting price raises or lowers revenue

Rule of thumb for revenue: cut price if demand is elastic, raise price if demand is inelastic, leave it alone if demand is unit-elastic.

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E_d = (% ΔQ_D) / (% ΔP)          TR-rule: d(TR)/dP has
                opposite sign to (|E_d| - 1)
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where: |E_d| > 1 elastic · |E_d| = 1 unit-elastic · |E_d| < 1 inelastic. Absolute value drops the minus sign implied by the law of demand.

WORKED EXAMPLE — ELASTIC DEMAND

A firm cuts price from 10.00 to 9.00 (−10%) and observes quantity sold rise from 100 to 120 (+20%). What is E_d, and what happens to revenue?

1. E_d = (+20%) / (−10%) = −2.0 ⇒ |E_d| = 2 > 1, demand is ELASTIC.
2. Old TR = 10 × 100 = 1,000; New TR = 9 × 120 = 1,080.
3. TR rose by 80 (+8%) because the quantity effect dominated the price effect.
4. If |E_d| had been 0.5 (inelastic), the same price cut would have LOWERED revenue.

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E_d = -2.0 → elastic → cut price, revenue rises.
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[1] By convention, E_d is reported as an absolute value in CFA materials — remember the minus sign for algebraic work.
[2] Elasticity is NOT the slope: slope is ΔQ/ΔP, elasticity is (ΔQ/Q)/(ΔP/P). A linear demand curve has constant slope but variable elasticity.

Source: CFA Institute, Curriculum Vol 2, Learning Module 1, demand/revenue discussion, p. 6.

Cross-price and income elasticities separate substitutes from complements and normal goods from inferior — the three-elasticity triad for any demand forecast

A demand forecast that doesn't specify the three elasticities is not a forecast — it is an assertion with no stress-test.

Elasticity	Formula	Sign	Interpretation & example
Own-price (E_d)	(% ΔQ_x) / (% ΔP_x)	Always < 0	Demand curve slopes down — gasoline E_d ≈ −0.3 (inelastic).
Cross-price, substitutes	(% ΔQ_x) / (% ΔP_y)	Positive	P_y up ⇒ Q_x up — Pepsi demand vs Coca-Cola price.
Cross-price, complements	(% ΔQ_x) / (% ΔP_y)	Negative	P_y up ⇒ Q_x down — printer demand vs ink cartridge price.
Income, normal good	(% ΔQ_x) / (% ΔI)	Positive	Income up ⇒ Q_x up — restaurant meals, airline tickets.
Income, luxury	(% ΔQ_x) / (% ΔI)	> 1	Share of wallet rises with income — fine wine, designer apparel.
			Income up ⇒ Q_x down — instant

WHAT SIGNALS A STRUCTURAL SHIFT

A rise in income elasticity for a staple (coffee becoming a "luxury" Nespresso SKU) or a sign-change in cross-elasticity (Netflix becoming a SUBSTITUTE for cinema) both signal that a category boundary is redrawing — and that incumbents need to re-price.

Source: CFA Institute, Curriculum Vol 2, Learning Module 1, demand analysis, pp. 6–7.

SECTION 02

Firm costs & profit maximization

Economic profit subtracts the opportunity cost of every factor — a firm with positive accounting profit can still be destroying economic value

Normal profit = zero economic profit = accounting profit equal to what the capital would have earned in its next-best use. It is a reward for participation, not a reward for excellence.

$$\text{Economic profit} = \text{TR} - [\text{Accounting costs} + \text{Opportunity costs}]$$

where: Accounting profit = TR − accounting costs. Economic profit = accounting profit − implicit opportunity costs.

THE ENTREPRENEUR EXAMPLE

Founder nets USD 120,000 in accounting profit from her consultancy. She had turned down a corporate role paying USD 150,000 to start the firm.

Accounting profit = +120,000.
Economic profit = 120,000 − 150,000 = −30,000.

She is destroying value and should return to corporate.

WHY IT MATTERS FOR INVESTORS

In a competitive long-run equilibrium, firms earn ZERO economic profit — equity returns just compensate for risk. Positive economic profit is a signal of a moat: a barrier protecting rents. The analyst's job is to assess whether the moat will hold.

Source: CFA Institute, Curriculum Vol 2, Learning Module 1, Overview, pp. 3–6.

A firm maximizes profit where marginal revenue equals marginal cost and marginal cost is rising — the single rule that governs every market structure

Under perfect competition, MR = P (horizontal demand). Under every imperfectly-competitive structure, MR < P (sloped demand). Rule unchanged.

$$\text{Maximize profit} \Leftrightarrow MR = MC \quad \text{AND} \quad dMC/dQ > 0$$

where: First-order condition: slope of TR equals slope of TC. Second-order condition: MC must cut MR from below.

CFA LM 1 EXAMPLE 1 — WR INTERNATIONAL, IMPERFECT COMPETITION

Prefab-housing firm facing a downward-sloping demand curve. Quantity Q moves in blocks of 10, price P falls 250 per block. At which Q does the firm break even, where is the profit region, and where is profit maximized?

1. Breakeven (TR = TC): Q = 40 at P = 9,000 → TR = TC = 360,000.
2. Region of profitability: 40 < Q < 80. At Q = 80, TR = TC = 640,000 (upper breakeven).
3. Profit maximum: Q = 60, P = 8,500 → TR = 510,000, TC = 480,000 → profit = 30,000.
4. Anywhere else in the profit region yields less than 30,000 because MR ≠ MC.

Q* = 60, P* = 8,500, max economic profit = 30,000.

Source: CFA Institute, Curriculum Vol 2, Learning Module 1, Example 1 (WR International), pp. 11–12.

Breakeven, shutdown, and long-run exit are three distinct thresholds defined by whether price covers total cost, variable cost only, or nothing

In the short run, fixed costs are sunk and only variable costs matter. In the long run every cost is variable, so the firm must cover ATC to justify staying.

Revenue vs cost relationship	Short-run decision	Long-run decision
TR ≥ TC (price ≥ ATC)	Operate — earning ≥ normal profit	Stay in market
TR = TC (price = min ATC)	Break-even point	Long-run equilibrium of perfect competition
TVC ≤ TR < TC (AVC ≤ P < ATC)	Operate — covers variable + part of fixed cost	Exit market when contract allows
TR = TVC (price = min AVC)	Shut-down point	Exit market
TR < TVC (price < AVC)	Shut down — minimize loss to TFC	Exit market

WHY ANALYSTS CHECK BREAKEVEN AND SHUTDOWN

For start-ups and turnarounds, knowing the unit-volume thresholds at which the firm (a) first stops losing money and (b) would rationally cease operations is more informative than a headline margin number. A firm operating just above its shutdown point is one demand-shock away from exit.

[1] Sunk fixed costs (non-avoidable) are ignored in the short-run decision — they will be paid whether the firm produces or not.

Source: CFA Institute, Curriculum Vol 2, Learning Module 1, "Breakeven Analysis and Shutdown", pp. 9–13.

Economies of scale pull long-run average cost down until minimum efficient scale; diseconomies push it back up — the classic U-shape of LRAC

LRAC is the envelope of all possible short-run ATC curves. Its minimum defines minimum efficient scale — the smallest plant size at which a firm is long-run competitive.

DRIVERS OF ECONOMIES OF SCALE

- Increasing returns to scale — output rises faster than inputs.
- Division of labor and specialization — each worker becomes proficient at a narrow task.
- Spread of fixed costs — high-CAPEX equipment amortized over more units.
- Bargaining power — bulk discounts on inputs.
- Efficient technology and learning-curve effects.

DRIVERS OF DISECONOMIES OF SCALE

- Bureaucracy and communication bottlenecks — decisions slow as hierarchy grows.
- Input-price pressure — bidding up wages, land, scarce talent.
- Overlap and duplication of business functions across divisions.
- Monitoring costs rise faster than output.
- Loss of founder / entrepreneurial culture.

Minimum efficient scale (MES) = output at which LRAC first reaches its minimum. Below MES a firm is sub-scale; far beyond it, diseconomies take over.

Source: CFA Institute, Curriculum Vol 2, Learning Module 1, "Economies & Diseconomies of Scale", pp. 14–19.

SECTION 03

Four market structures

The four market structures differ on number of sellers, product differentiation, barriers to entry, and pricing power — one grid summarizes all four

Pricing power rises, quantity supplied falls, and consumer welfare falls as you move from perfect competition toward monopoly. Economic profit in the long run is zero on the left and positive on the right.

Structure	Sellers	Product	Barriers to entry	Pricing power	Long-run econ. profit
Perfect competition	Many	Homogeneous	Very low	None	0
Monopolistic competition	Many	Differentiated	Low	Some	0
Oligopoly	Few	Homogeneous or diff.	High	Some / considerable	> 0 possible
Monopoly	One	Unique	Very high	Considerable	> 0

PORTER'S FIVE FORCES — THE PRACTITIONER'S OVERLAY

Academic theory asks "how many sellers?"; Porter asks five practical questions: threat of entry, power of suppliers, power of buyers, threat of substitutes, and rivalry among incumbents. Four of the five map directly onto the market-structure grid. The fifth — power of suppliers — is the one academic economics under-weights and that practitioners use to identify durable moats ("economic moats").

Source: CFA Institute, Curriculum Vol 2, Learning Module 1, Exhibit 13, pp. 23–24.

In perfect competition price equals marginal cost and economic profit is zero in the long run — the benchmark every other structure is measured against

No firm has pricing power, so the demand curve each firm faces is perfectly elastic (horizontal) at the market price — P = MR = AR.

Short run: P = MR = MC Long run: P = MR = MC = min(ATC)

where: Free entry drives long-run price down to minimum ATC, at which point economic profit is zero (firms earn only normal profit).

INDUSTRY FEATURES

- Many small sellers, each a price taker.
- Identical, undifferentiated output.
- Perfect information.
- No entry or exit costs.

SHORT-RUN ECON. PROFIT

Possible if the firm's ATC is below the market price. But the excess attracts entrants, and over time they drive market price down to every firm's min ATC.

REAL-WORLD PROXIES

Commodity agriculture (corn, wheat), spot FX between major pairs, electronically-traded large-cap equities. Few pure examples — but a useful idealization.

Source: CFA Institute, Curriculum Vol 2, Learning Module 1, Perfect Competition, pp. 7–9.

Monopolistic competition combines many sellers with product differentiation — firms face downward-sloping demand but still earn zero long-run economic profit

Differentiation gives short-run pricing power; low barriers to entry eventually erode the margin — the hybrid structure between competition and monopoly.

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Short run:   choose Q so MR = MC, charge P from demand curve   ⇒
                profit  =  (P − ATC) × Q
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where: In long-run equilibrium, entry drives demand curve down until it is tangent to ATC — profit = 0 but output still below MES.

WHY IT DOMINATES REAL MARKETS

Most consumer markets are monopolistically competitive: soft drinks, cosmetics, apparel, cafés. Differentiation supports margin; variety pays for the slight inefficiency. Schumpeter: differentiation is the profit incentive to innovate.

THE BRAND MOAT

Successful differentiation — Coca-Cola, Harley-Davidson — makes the firm look more monopoly-like, sustaining margins longer. Analysts should track brand health surveys and advertising ROIs, not just price elasticity.

Source: CFA Institute, Curriculum Vol 2, Learning Module 1, Monopolistic Competition, pp. 25–28.

The Cournot duopoly assumes each firm treats its rival's output as fixed — solving the two reaction curves gives a price between competition and monopoly

Cournot (1838) predates game theory but gives the same answer as the Nash equilibrium of the output-choice game — each firm best-responds to the rival's quantity.

Setup: $Q_D = 450 - P$, $MC = 30$, $Q_D = q_1 + q_2$

where: Each firm maximizes its profit assuming the rival's quantity is fixed. Solve simultaneously.

CFA LM 1 — DUOPOLY WITH LINEAR DEMAND

Two firms choose quantities q_1 and q_2. $P = 450 - (q_1+q_2)$. Each sets MR = MC = 30 taking the other's q as given.

1. Firm 1: $TR_1 = (450 - q_1 - q_2)\cdot q_1 \rightarrow MR_1 = 450 - 2q_1 - q_2 = 30$.
2. By symmetry: $450 - q_1 - 2q_2 = 30$. Solve the system: $q_1 = q_2 = 140$, $Q = 280$.
3. Price: $P_{Cournot} = 450 - 280 = 170$.
4. Benchmarks: perfect competition (P = MC) → Q = 420, P = 30. Monopoly (MR = MC on market demand) → Q = 210, P = 240.

Cournot: Q = 280, P = 170 — squarely between competition (30) and monopoly (240).

Source: CFA Institute, Curriculum Vol 2, Learning Module 1, "The Cournot Assumption", pp. 32–33.

In a Nash equilibrium no firm can profitably deviate given rivals' strategies — but the collective best outcome often needs collusion each player is tempted to break

The Nash outcome maximizes each player's own profit given rivals' best responses. The collusive outcome maximizes joint profit — but requires trust and enforcement.

ArcCo strategy \ BatCo strategy	BatCo — LOW price	BatCo — HIGH price
ArcCo — LOW price	ArcCo 50 · BatCo 70	ArcCo 80 · BatCo 0
ArcCo — HIGH price	ArcCo 300 · BatCo 350	ArcCo 500 · BatCo 300

READING THE PAYOFF MATRIX

Each firm picks Low or High simultaneously. Payoffs are (ArcCo, BatCo). Find the Nash equilibrium and the cooperative equilibrium.

1. Given ArcCo HIGH: BatCo prefers LOW (350 > 300). So (High, High) is NOT a Nash equilibrium.
2. Given ArcCo LOW: BatCo prefers LOW (70 > 0). Given BatCo LOW: ArcCo prefers LOW (50 > —, ArcCo weakly prefers Low given BatCo Low).
3. Nash equilibrium: (Low, Low) with payoffs 50 + 70 = 120 — a prisoners'-dilemma outcome.
4. Cooperative outcome (High, High) pays 500 + 300 = 800 in total but requires ArcCo to share some of its 500 with BatCo to keep BatCo from defecting.

Nash = (Low, Low). Cooperative = (High, High). Gap = 800 – 120 = 680 in foregone joint profit.

[1] Formal collusion is illegal in most jurisdictions (antitrust law). Implicit collusion via signals is harder to detect but still attracts regulatory attention.

Source: CFA Institute, Curriculum Vol 2, Learning Module 1, "The Nash Equilibrium" (ArcCo / BatCo), pp. 34–36.

A single-price monopolist sets MR equal to MC on the whole market demand curve — producing less and pricing higher than competition

The monopoly price is the one the demand curve supports at MR = MC. The gap between competitive and monopoly price is the deadweight loss society pays for the monopolist's rent.

Same setup: $P = 450 - Q$, $MC = 30$. Monopolist solves MR = MC on the full market demand.

where: $TR = P \cdot Q = (450 - Q) \cdot Q \rightarrow MR = 450 - 2Q$. Set $MR = MC$: $450 - 2Q = 30 \rightarrow Q = 210$; then $P = 450 - 210 = 240$.

Market structure	Quantity	Price	Consumer gain	Producer surplus
Perfect competition (P = MC)	420	30	Maximum	Zero econ. profit
Cournot duopoly (two firms)	280	170	Intermediate	Positive

PRICE DISCRIMINATION — THE MONOPOLIST'S UPGRADE

If the monopolist can charge different prices to different buyers (airlines, software licensing) it captures more of the consumer surplus as profit. First-degree (perfect) discrimination eliminates deadweight loss but requires perfect information — rare. Third-degree (segment-based) is common: student pricing, geographic pricing, volume tiers.

Source: CFA Institute, Curriculum Vol 2, Learning Module 1, monopoly vs competition comparison, pp. 32–33.

Concentration ratios and the Herfindahl-Hirschman Index measure how concentrated a market is — and the HHI catches mergers the CR ignores

HHI squares market shares before summing — a market of five equal firms has HHI = 0.20; a monopolist has HHI = 1.00.

$$CR_N = \Sigma_{i=1..N} s_i \qquad HHI = \Sigma_i s_i^2$$

where: s_i = market share of firm i (as a decimal). CR_N uses the top N firms; HHI typically uses all or top 50.

CFA LM 1 — FIVE-FIRM MARKET

Five firms with shares: Zeta 35%, Yusef 25%, Xenon 20%, Waters 10%, Vlastos 10%. Compute CR_3 and HHI_3.

1. CR_3 = 35% + 25% + 20% = 80% — top three control 80% of the market.
2. $HHI_3 = 0.35^2 + 0.25^2 + 0.20^2 = 0.1225 + 0.0625 + 0.0400 = 0.2250$.
3. Merger scenario: Zeta + Yusef merge to 60%. CR_3 only rises to 0.60 + 0.20 + 0.10 = 90% (+10 pp).
4. HHI_3 jumps to $0.60^2 + 0.20^2 + 0.10^2 = 0.36 + 0.04 + 0.01 = 0.41$ — the merger DOUBLES HHI but barely moves CR.

Pre-merger HHI_3 = 0.225. Post-merger HHI_3 = 0.41 — HHI is the more merger-sensitive gauge.

[1] US DOJ/FTC Horizontal Merger Guidelines classify HHI < 0.15 as unconcentrated; 0.15–0.25 moderately concentrated; > 0.25 highly concentrated.

Source: CFA Institute, Curriculum Vol 2, Learning Module 1, "Determining Market Structure", pp. 41–43.

SECTION 04

Business cycle phases & indicators

The business cycle has four phases — recovery, expansion, slowdown, contraction — distinguished by the output gap relative to potential GDP

The output gap = actual GDP minus potential GDP. Negative in recovery and contraction; positive in expansion and slowdown. Rate-of-change of the gap defines the phase.

Phase	Output gap	Employment	Inflation
Recovery	Negative, narrowing	Layoffs slow, hiring resumes	Moderate
Expansion	Negative → positive	Hiring accelerates, U falls	Picks up modestly
Slowdown	Positive, narrowing	Hiring continues but decelerates	Accelerates further
Contraction	Positive → negative	Hours cut, then layoffs	Decelerates with a lag

RULE OF THUMB VS NBER METHOD

The informal rule: two consecutive quarters of negative real GDP growth = recession. The NBER method looks at a broader basket — employment, industrial production, real personal income, manufacturing and trade sales — and dates business cycles with a lag of months or years. The simple rule can mislead (a slightly-positive middle quarter between two negatives wouldn't count) but is operationally useful for real-time calls.

Source: CFA Institute, Curriculum Vol 2, Learning Module 2, "Phases of the Business Cycle", pp. 52–55.

Classical, growth, and growth-rate cycles are three different measuring sticks — most modern analysts use the growth cycle around trend GDP

The classical cycle is rarely useful today because true level-declines in GDP are uncommon. The growth cycle speaks the language of most modern macro.

CLASSICAL CYCLE

Fluctuations in the LEVEL of economic activity (volume GDP).

Expansions are long; contractions short. Peaks = local maxima, troughs = local minima of the level.

Rarely used in practice — GDP falls in absolute terms less than once a decade in most developed economies.

GROWTH CYCLE

Fluctuations AROUND the trend / potential growth path — the output gap.

Peaks reached earlier, troughs later than in the classical cycle.

Most common framework today. Used by OECD, IMF, Fed — and by this deck.

GROWTH-RATE CYCLE

Fluctuations in the GROWTH RATE itself (e.g., YoY GDP growth).

No need to estimate trend. Peaks and troughs identified earliest of the three.

Matches how central-bank watchers phrase calls: "growth has peaked."

Source: CFA Institute, Curriculum Vol 2, Learning Module 2, "Types of Cycles", pp. 52–54.

Credit cycles lead business cycles and amplify them — loose credit seeds the next recession by funding the excesses that later unwind

Average credit cycle is longer than average business cycle. Recessions paired with a credit bust (housing, leverage) are deeper and longer than those without.

WHAT CREDIT CYCLES MEASURE

The changing AVAILABILITY and PRICING of credit. Proxies: private credit / GDP, real house prices, commercial bank lending standards, high-yield spread, LTV ratios on new mortgages.

WHY THEY AMPLIFY

Loose credit funds construction and real-estate purchases. Bubbles in asset prices feed back into household wealth and collateral values, inflating the boom. When credit tightens, the unwind is self-reinforcing — defaults → bank losses → further tightening.

Crisis	Region	Credit-cycle precursor
Latin America debt crisis, 1980s	Latam	Sovereign dollar borrowing boom + commodity bust
Asian financial crisis, 1997–1998	East Asia	Property and corporate USD-lending boom
Global Financial Crisis, 2008–2009	Global	US subprime mortgage leverage + securitization
Euro-area sovereign crisis, 2010–2012	Eurozone	Periphery bank-sovereign doom loop

Leading, coincident, and lagging indicators each answer a different question — what will happen, what is happening, what just happened

A good phase call uses all three together: a coincident indicator to anchor the present, a leading one to extrapolate, and a lagging one to confirm the last turn.

Category	Example indicators (US)	Use
Leading	Weekly hours (mfg), initial jobless claims, mfg new orders, building permits, S&P 500, yield-curve slope, consumer expectations	Predict the next turn
Coincident	Industrial production index, real personal income, manufacturing and trade sales, non-farm payrolls	Identify the current phase
Lagging	Average duration of unemployment, inventory-to-sales ratio, unit labor cost, average prime rate, consumer instalment debt / income	Confirm the previous turn

DIFFUSION INDEX — READING BREADTH, NOT JUST LEVEL

A diffusion index assigns +1 / 0.5 / 0 to each constituent of a composite based on whether it rose, stayed flat, or fell — then averages and scales to 100. Breadth matters: a composite rising because 8 of 10 components are up is a stronger signal than one dragged up by two outliers.

Source: CFA Institute, Curriculum Vol 2, Learning Module 2, "Economic Indicators", Exhibit 10–11, pp. 66–67.

SECTION 05

Unemployment & inflation

The unemployment rate counts the jobless who are actively looking for work — on a labor-force base that itself shrinks in downturns

Unemployment is a lagging indicator because businesses delay both hiring (until recovery is confirmed) and firing (because training costs are sunk). This is why recoveries often feel "jobless" at first.

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U = (Unemployed / Labor force)        LFPR = Labor force / Working-age population
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where: Labor force = employed + unemployed (actively seeking). Excludes discouraged workers and the institutionalized population.

Component	Definition	Cycle behavior
Frictional	Short-term job transitions, search friction between positions	Roughly stable
Structural	Skill / geographic mismatch between vacancies and the workforce	Slow-moving
Cyclical	Driven by aggregate demand — rises in recession, falls in expansion	Highly cyclical

Subtle point: the headline U-rate can FALL early in a recession because discouraged workers drop out of the labor force — check LFPR alongside U.

Source: Illustrative — standard Level I macroeconomics framework; consistent with CFA Vol 2, LM 2 cycle commentary.

CPI, core CPI, PPI, and the GDP deflator measure inflation differently — and they can disagree by meaningful margins

Central banks target core CPI or core PCE because food and energy are volatile supply-driven components that monetary policy cannot address without wrecking employment.

Measure	Basket / scope	Weights	Typical use
Headline CPI	Urban household consumption basket	Fixed-weight (Laspeyres)	Cost-of-living indexation
Core CPI	CPI excluding food and energy	Same as CPI ex-volatiles	Central-bank target
PPI	Producer selling prices at wholesale	Industry output-weighted	Early-stage inflation signal
GDP deflator	All domestically produced goods and services	Current-period weights (Paasche)	Whole-economy price measure

WHY HEADLINE AND CORE CAN DIVERGE

An oil-price spike lifts headline CPI immediately but raises core CPI only to the extent it leaks into transport, heating, and input costs of non-energy goods. The Fed sees through the first effect and reacts only to the second — the "look-through" principle. This is why stories like "CPI print high, Fed unmoved" are not contradictions.

Source: Illustrative — standard Level I macroeconomics framework; consistent with central-bank practice per CFA Vol 2, LM 4.

The Phillips curve maps the short-run trade-off between inflation and unemployment — but the long-run curve is vertical at the natural rate

Short-run Phillips curve: the central bank can buy less unemployment by accepting more inflation. Long-run: it cannot — inflation rises with no gain in employment.

$$\text{Short-run:} \quad \pi = E[\pi] - \beta\,(U - U_{natural}) \qquad \text{Long-run:} \quad U = U_{natural} \quad \text{(vertical)}$$

where: π = inflation; $E[\pi]$ = expected inflation; U = unemployment rate; $U_{natural}$ = NAIRU; $\beta > 0$ is the short-run slope.

SHORT RUN

Given fixed expectations, U and π trade off. A demand surprise that pushes U below NAIRU raises π above E[π]. Central banks exploit this briefly to "run the economy hot."

LONG RUN

Agents update expectations. If actual π > expected π every period, E[π] rises and the short-run curve shifts up — same U, higher π. At U = NAIRU the long-run curve is vertical.

This is the Friedman-Phelps result.

Source: Illustrative — standard Level I macroeconomics; Phillips (1958), Friedman–Phelps expectations-augmented extension.

Okun's law ties the output gap to the unemployment gap at roughly 2-to-1 — a rule of thumb that holds across most developed economies

Okun's law is the bridge from GDP growth forecasts to unemployment forecasts — and, combined with Phillips, to inflation forecasts.

$$(Y - Y^*) / Y^* \approx -\alpha \times (U - U_{natural}) \quad \text{(developed markets)} \qquad \alpha \approx 2.0$$

where: A 1 pp rise in the unemployment rate above NAIRU corresponds to a ~2% output gap below potential.

ILLUSTRATIVE — MAPPING U-GAP TO OUTPUT GAP

Economy with potential growth 2% p.a. NAIRU = 4%. Observed unemployment rises from 4% to 6% over the year. Estimate the output gap.

1. Unemployment gap: U − U_natural = 6% − 4% = +2 pp.
2. Okun with α = 2.0: output gap ≈ −2.0 × 2 pp = −4%.
3. Interpretation: actual GDP is running ~4% below potential — consistent with a recession.
4. Reverse use: a 3% recovery in output closes about 1.5 pp of the unemployment gap.

U-gap +2 pp → output gap ≈ −4% (Okun coefficient 2.0).

[1] The coefficient varies by country and era. Post-1990 US: closer to 2.0. Japan: closer to 4–6 (lifetime employment damps the relationship).

Source: Illustrative — Okun (1962) original coefficient ≈ 2.0 for the US; 1.5–2.5 range across developed markets.

Combining Phillips and Okun lets an analyst translate an unemployment forecast into an inflation forecast — a core input for rate-call narratives

This is the quick-and-dirty model central-bank watchers run in their heads: output gap ⇒unemployment gap ⇒inflation gap.

ILLUSTRATIVE — RECESSION SCENARIO

Starting state: U = U_natural = 4%, π = E[π] = 2%. Shock: demand-driven recession pushes U to 6%. Okun α = 2.0, Phillips β = 0.5. Work through the implications.

1. Step 1 — Unemployment gap: U − U_natural = +2 pp.
2. Step 2 — Okun output gap: −2.0 × 2 pp = −4% (actual GDP 4% below potential).
3. Step 3 — Phillips: π = E[π] − β × (U − U_natural) = 2% − 0.5 × 2 = 1%.
4. Step 4 — Central-bank reading: 100 bp of inflation undershoot + 4% output gap = case for expansionary policy.

π falls from 2% to 1% via the short-run Phillips trade-off; output gap = −4%.

WHERE THIS MODEL BREAKS

The short-run Phillips slope β has flattened materially post-1990s (well-anchored inflation expectations), so inflation responds less to U-gap changes than older textbooks imply. Supply shocks (energy, pandemic) move π and U in the SAME direction and break the trade-off entirely — which is why 2021–2023 stagflation confounded the simple model.

Source: Illustrative — applies the relationships from slides 29–30 to a stylized scenario.

SECTION 06

Investment-relevant cycle dynamics

Equity markets lead the real economy by three-to-six months while credit spreads and the yield-curve slope lead further still — both live in the LEI

The equity market's job is to discount future cash flows — so it turns before reported earnings. Credit markets discount future default probability — they turn first.

EQUITY MARKETS

Typically bottom 3–6 months before the real economy troughs. Top 3–6 months before the peak.

S&P 500 is one of the 10 Conference Board LEI components for this reason.

Take-away: by the time GDP confirms a turn, equities have already priced it.

YIELD CURVE

10y–2y or 10y–fed-funds spread.

Inversion has preceded each US recession since 1970 by 12–24 months (one false signal in the mid-1960s).

Mechanism: short rates expected to fall below long rates when the market expects a rate-cut cycle to defend against recession.

CREDIT SPREADS

High-yield spread over Treasuries widens before default cycles.

Spreads tend to trough mid-cycle and rise through the late cycle as underwriting deteriorates.

Leading Credit Index sits in the LEI.

Source: CFA Institute, Curriculum Vol 2, Learning Module 2, "Market Conditions and Investor Behavior" + LEI, pp. 55–67.

Inventory and capital-spending cycles amplify the real economy — orders for light equipment move first, heavy capex last

Inventories are small as a share of GDP but swing violently. Capex is the single most pro-cyclical component of GDP — it's what makes cycles big.

Phase	Inventory behavior	Capex behavior	Typical orders
Recovery	I/S ratio stable as sales recover	Low but rising; focus on efficiency, not capacity	Software, IT hardware
Expansion	Inventory rebuild; rising I/S briefly	Capacity expansion; strong cash flow supports capex	Heavy equipment, factories
Slowdown	I/S stable-to-rising as sales decelerate	New-order growth slows; late-cycle capacity still being added	Construction, infrastructure
Contraction	I/S spikes as production overshoots sales	New orders halted; some in-flight orders canceled	Maintenance scaled back

WHY THIS MATTERS FOR SECTOR SELECTION

Industrials and semiconductors often lead early-cycle because their customers restart equipment orders first. Construction and aerospace lead late-cycle because long-planning projects carry on even as the broader economy rolls over. Any sector view should ask: "where does this sit on the order-book lead-lag curve?"

Source: CFA Institute, Curriculum Vol 2, Learning Module 2, Exhibits 8–9, pp. 61–64.

Sector rotation maps cycle phases to portfolio tilts — early-cycle favors consumer discretionary, mid-cycle industrials, late-cycle energy, and recessions staples and utilities

The playbook: risk-on assets (small caps, cyclicals, HY credit) lead early; defensives (staples, utilities, long duration) lead late and in contraction.

Phase	Equity style / size	Favored sectors	Fixed income
Recovery	Small caps, value, cyclicals	Consumer discretionary, financials, homebuilders	HY credit rally
Expansion	Mid-cap quality, growth	Industrials, technology, materials	IG credit tight
Slowdown	Large-cap quality, low volatility	Energy, late-cycle industrials, commodities	Duration extends
Contraction	Defensives, long duration bonds	Consumer staples, utilities, healthcare	Government bonds

LIMITATIONS — DO NOT MISTAKE THIS FOR A FORECAST

Sector rotation works on average and in hindsight. Every cycle has its own character — technology led both the late-1990s expansion and the 2020–2021 recovery; energy led the 2022 slowdown but underperformed most previous ones. Use the playbook as a prior, anchor it to the current cycle's specific drivers, and size bets with humility.

[1] Sector playbooks are empirical averages over many cycles, not forecasts. Use them as a prior — then overlay the specific drivers of the current cycle.

Source: CFA Institute, Curriculum Vol 2, Learning Module 2, "Market Conditions and Investor Behavior", pp. 55–57.

KEY TAKEAWAYS

Deck 4 — Microeconomics & Business Cycles

- Own-price elasticity $E_d = (\%\Delta Q)/(\%\Delta P)$; $|E_d| > 1$ elastic, < 1 inelastic. Cut price when elastic to raise revenue; raise price when inelastic.
- Breakeven at $P = \min(ATC)$; short-run shutdown at $P < \min(AVC)$; long-run exit when $P < \min(ATC)$. Sunk fixed costs are irrelevant to the short-run choice.
- CFA Cournot duopoly ($Q = 450 - P$, $MC = 30$): $P_{competition} = 30$, $P_{Cournot} = 170$, $P_{monopoly} = 240$. Nash (ArcCo/BatCo) landed at Low/Low, foregoing 680 of joint profit.
- Leading indicators predict, coincident ones describe, lagging ones confirm. Equity and yield-curve live in the LEI; unemployment and inventory-sales live in the lagging set.
- Every market structure maximizes profit at $MR = MC$ with MC rising. Under perfect competition $P = MR$; under imperfect competition $MR < P$.
- Four structures — perfect, monopolistic, oligopoly, monopoly — differ on sellers, differentiation, barriers, pricing power. HHI detects concentration better than the concentration ratio.
- Four business-cycle phases (recovery, expansion, slowdown, contraction) defined by the output gap. Credit cycles lead and amplify business cycles.
- Phillips (short-run $\pi = E[\pi] - \beta(U - U^*)$) combined with Okun (output gap $\approx -2 \times$ U-gap) maps an unemployment forecast to an inflation forecast — core input for any rate call.